OPENNESS OF VERSALITY FROM A PERFECT OBSTRUCTION THEORY INPUT

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 4363-4385 of sources/stacks-artin/pinned-original-source.tex; proof span is lines 4387-4402, the complete author proof (14 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 42 of queue_artin.json (pb=631, stmt 4363-4385, proof 4387-4402).

Editorial note. stacks-artin wave ar01 (2026-10-14): fresh adjudication accept per lane-s3/artin_judgments.md; pool per Chief 10:5x direct dispatch (idle-first over S12's 01:5x note, SMC precedent); artin.tex 99 rows/67 with proof judged 46 accept / 21 reject / 5 AGP-overlap avoidance rows; patch-sd-envs contract spans scanned.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 07YM: sources/stacks-artin/pinned-original-source.tex, source lines 4235–4251. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 068U: sources/stacks-artin/pinned-original-source.tex, source lines 4402–4402. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 07YC: sources/stacks-artin/pinned-original-source.tex, source lines 3806–3838. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a locally Noetherian scheme. Let $\mathcal{X}$ be a category fibred in groupoids over $(Sch/S)_{fppf}$ satisfying (RS*). Let $U = \mathrm{Spec}(A)$ be an affine scheme of finite type over S which maps into an affine open $\mathrm{Spec}(\Lambda)$. Let x be an object of $\mathcal{X}$ over U. Let $\xi : E \to NL_{A/\Lambda}$ be a morphism of $D^-(A)$. Assume*

(i) *for every deformation situation $(x, A' \to A)$ we have: x lifts to $\mathrm{Spec}(A')$ if and only if $E \to NL_{A/\Lambda} \to NL_{A/A'}$ is zero,*

Date: stacks-project:artin.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

(ii) *there is an isomorphism of functors* $T_x(-) \to \mathrm{Ext}^0_A(E, -)$ *such that* $E \to NL_{A/\Lambda} \to \Omega^1_{A/\Lambda}$ *corresponds to the canonical element (see Remark 07YC),*
(iii) *the cohomology groups of* E *are finite* A*-modules.*

If x *is versal at a closed point* $u_0 \in U$*, then there exists an open neighbourhood* $u_0 \in U' \subset U$ *such that* x *is versal at every finite type point of* U'*.*

Proof. Let C be the cone of ξ so that we have a distinguished triangle

$$E \to NL_{A/\Lambda} \to C \to E[1]$$

in $D^-(A)$. By Lemma 07YM the assumption that x is versal at u_0 implies that $H^{-1}(C \otimes^{\mathbf{L}} k) = 0$. By More on Algebra, Lemma 068U there exists an $f \in A$ not contained in the prime corresponding to u_0 such that $H^{-1}(C \otimes^{\mathbf{L}}_A M) = 0$ for any A_f-module M. Using Lemma 07YM again we see that we have versality for all finite type points of the open $D(f) \subset U$. □